

SkillCall

One free phone call. Right skill. Right job.



Works on any keypad phone



Talks in 22 Indian languages



Runs 100% on the cloud

PROBLEM STATEMENT ID

SIH26097

PROBLEM STATEMENT TITLE

AI-driven voice assistant for livelihood mapping and NSQF-aligned skilling recommendations for SC communities under the GIA component of PM-AJAY

THEME

Agriculture, FoodTech & Rural Development

PS CATEGORY

Software

TEAM ID

_____ **(fill from the SIH portal)**

TEAM NAME (REGISTERED ON PORTAL)

Cognify

Proposed Solution · one free phone call that turns a person's skills into the right work



THE PROBLEM PM-AJAY pays for skill training and business support for poor SC families — but paper forms and field officers pick the course, so people who can't read or don't know the trades land in the wrong one.

41%

of trained people got a job

CAG 2025

34%

of SC adults cannot read

Census 2011

36%

of women have no own phone

NFHS-6

HOW ONE CALL WORKS



1

Missed call

Free call-back in his own language within a minute.



2

Basics by keypad

Age, education, distance by keys — a speech error can't change eligibility.



3

He talks, AI listens

Says what work he does; AI asks follow-up skill questions.



4

Read back & saved

Profile read back, press 1 to confirm, stored as one table row.



5

1-4 verified options

Explained aloud with place, time, benefit. He can ask anything.



6

Choose → CSC → officer

SMS Case ID; documents at CSC; officer approves; follow-up calls at 1/3/6 months.

WHAT'S NEW — 8 THINGS A FORM CAN NEVER DO



Any keypad phone

No app, no internet, no reading.



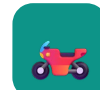
Ask anything

Every doubt answered by AI in his language.



Tables, not forms

Officers only review ready, arranged data.



Smart questions

Bike mechanic → engine? tyres? oil service?



Closed knowledge

Answers only from govt data we load — never the internet.



Validator AI (SATYA)

Re-checks every fact before it is spoken.



Cheapest model first

Bigger model, then a human — only if needed.



Web portal too

Mobile + OTP login; edit anything, anytime.

TECH STACK



Calls



Exotel



WhatsApp API



AI (Indian, open-source)

Sarasv AI • Saaras,
Bulbul, 30B / 105BAI4Bharat
IndicConformer

Pipecat voice engine



Bhashini (MeitY)



Backend



Python



FastAPI



Celery



Redis



Data & apps



PostgreSQL + pgvector



React portal



Docker



MeghRaj govt cloud*

* needs government MoU. Logos: official marks (Devicon, Simple Icons, project GitHub avatars).

HOW IT WORKS

LIVE CALL — every sentence goes round this loop in ~1.5 s

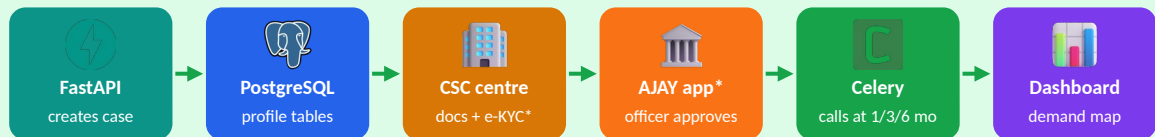


Router: free model first → paid / bigger model only if unsure or the caller is unhappy → human last.



Knowledge base: govt rules, NQR courses, NCO jobs, centres — the only source of facts. No internet.

AFTER THE CALL — the confirmed profile becomes a case



Web portal — login with mobile + OTP. Family, CSC or officer can view / edit the same data anytime, or do the whole thing on the web.



Stored as tables — profiles · skills · options · cases · follow-ups · audit log. No Aadhaar number, audio deleted after confirmation.

AI understands → verified data decides → SATYA checks → humans approve

[Architecture details, API docs and code → GitHub repo \(add link\)](#)

Can it be built and run? Yes — every piece already works in India today, and one completed call costs about ₹37–83.

ALREADY PROVEN IN INDIA



10 of 22 languages

UIDAI runs Sarvam's voice AI for Aadhaar today

PIB / MeitY 2025



12 lakh calls / day

Kilkari, the govt's health-call system

BMJ Global Health 2021



≤ 20% word error

best Indian speech AI on real phone calls

Voice of India 2026



All APIs live now

Sarvam, Pipecat, Exotel, PostgreSQL — build today

our prototype plan

WHAT ONE COMPLETED CALL COSTS

Phone line (Exotel)	18 min × ₹0.80–1.80	₹17–38
Speech ↔ text (Saaras, Bulbul)	₹0.50 / min + ₹3 / 1,000 chars	₹12–16
AI brain (Sarvam-30B / 105B)	≤ ₹73 per 1M tokens	₹1–3
SMS + human help + CSC fee	only 10–40% of callers need help	₹7–24
Hosting & numbers	spread per profile	₹1–3
TOTAL per completed profile	1,000 missed calls → 517 finished profiles	₹37–83

Unit prices from Sarvam and Exotel pricing pages; the funnel (76% answer × 80% finish × 85% complete) is our estimate.

MONEY CHECK



0.04–0.08% of the budget

₹0.9–1.7 crore a year runs a whole state; PM-AJAY has ₹2,140 crore.



₹9–13 lakh pilot

1 block, 1,000 families, 6 months — then scale.



6.5 lakh CSCs ready

+ 90 lakh SHGs and the AJAY app already exist — we plug in, not rebuild.

RISKS WE PLANNED FOR

Village dialects → facts by keypad · every answer read back · bigger model, then a human

Unknown numbers ignored → he calls first; call-back from an official 1600 number

AI says something wrong → SATYA blocks anything not in the govt database

No phone (36% of women) → register at the nearest CSC or on the web portal

Who gains? Poor SC families, CSC operators, district officers and the Ministry. Every number shows its working.

WHAT CHANGES



More people get jobs

41% → 70%

Only 41% of trained people get a job today; the govt target is 70%. That gap is what we close.



People stay in jobs

+11 points

Trainees told the truth about the job before joining were 11 points more likely to stay (field trial, 2024).



Officers' time back

~24 staff-years / yr

50 minutes saved per person × 50,000 GIA beneficiaries a year.



Money not wasted

₹12.5 crore / yr

If just 5% of wrong matches are avoided: 2,500 people × ₹50,000 support.

WHO WINS



SC family

Right course the first time, free, in his language — and a follow-up until he earns.



CSC operator

Case arrives pre-filled; only documents and e-KYC left to do.



District officer

Ready, checked cases in the AJAY app + a demand map for the annual plan.



Ministry

One record from call to job; outcomes marked confirmed or self-reported.

BENEFITS



Social

Women and non-readers included; no reading, no app.



Economic

Less money on wrong training; more people earning, proved by follow-up.



Governance

One shared case record; planning data instead of paper forms.



Environment

No paper, fewer office trips, shared govt cloud.



Government data

- CAG Report No. 20 of 2025 — PMKVY audit (41% placed) cag.gov.in
- PM-AJAY GIA norms & beneficiaries — PIB 2023 / 2025 pib.gov.in
- PM-AJAY budget ₹2,140 cr — Demands for Grants 2026-27 indiabudget.gov.in
- NFHS-6 (women's phone access) · Census 2011 (SC literacy) pib.gov.in



Technology

- Sarvam AI — Saaras v3, Bulbul v3, Sarvam-30B / 105B, pricing sarvam.ai
- AI4Bharat IndicConformer — open Indian speech recognition ai4bharat.iitm.ac.in
- Pipecat — open-source voice-agent framework pipecat.ai
- Bhashini (MeitY) · Exotel · PostgreSQL · FastAPI · React bhashini.gov.in



Research papers

- Chakravorty et al. 2024, J. Dev. Econ. — honest job info → +11 pts retention [sciencedirect.com](https://www.sciencedirect.com)
- Bashingwa et al. 2021, BMJ Global Health — Kilhari at 12 lakh calls/day gh.bmi.com
- Voice of India 2026 — Indian speech-AI benchmark on real calls arxiv.org
- Medhi et al. 2011, ACM TOCHI — voice UIs for low-literacy users dl.acm.org



Standards & policy

- NCO-2015 occupation codes · NQR / NCVET job roles nqr.gov.in
- DPDP Rules 2025 — consent and data protection meitv.gov.in
- TRAI 1600-series rules for government calls (2026) traigov.in
- UIDAI-Sarvam voice AI deployment — PIB / MeitY 2025 pib.gov.in



See more

code, working prototype, demo video and the full 16-part reference file with every calculation

[GitHub repo →](#)[Live prototype →](#)[Demo video →](#)[Reference file →](#)